SRI: N03LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 3$. The Hubble-flow parameters are denoted by ϵ_j and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1}\ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

1 Spectra

1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*}} \left[\sum_{d=0}^{3} F_{s,d} + O(\mathrm{SR}^4)\right].$$

Relative order 0.

$$F_{s,0} = 1.0$$

Relative order 1.

$$F_{s,1} = 0.72963715453852\epsilon_2 - 0.54072569092296\epsilon_1$$

Relative order 2.

$$\begin{aligned} F_{s,2} = {} & 1.5402686460338\epsilon_1^2 + 0.49988573877771\epsilon_2^2 \\ & + 1.0192767657904\epsilon_1\epsilon_2 + 0.14504832807052\epsilon_2\epsilon_3 \end{aligned}$$

Relative order 3.

$$\begin{aligned} F_{s,3} = {} & 0.20072294544592\epsilon_2^3 + 1.6057835635673\epsilon_1^3 \\ & + 4.2977813647689\epsilon_1^2\epsilon_2 + 1.959600275274\epsilon_1\epsilon_2^2 \\ & + 0.030669313835717\epsilon_2\epsilon_3^2 - 0.23520055946694\epsilon_2^2\epsilon_3 \\ & + 0.030669313835717\epsilon_2\epsilon_3\epsilon_4 - 0.65969152604436\epsilon_1\epsilon_2\epsilon_3 \end{aligned}$$

1.2 Tensor spectrum

$$P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2} \left[\sum_{d=0}^{3} F_{t,d} + O(\mathrm{SR}^4)\right].$$

Relative order 0.

$$F_{t,0} = 1.0$$

Relative order 1.

$$F_{t,1} = -0.54072569092296\epsilon_1$$

Relative order 2.

$$F_{t,2} = 1.5402686460338\epsilon_1^2 - 0.25062903478191\epsilon_1\epsilon_2$$

Relative order 3.

$$\begin{aligned} F_{t,3} = {} & 1.6057835635673\epsilon_1^3 + 1.8891060194179\epsilon_1^2\epsilon_2 \\ & - 0.18929040711048\epsilon_1\epsilon_2^2 - 0.18929040711048\epsilon_1\epsilon_2\epsilon_3 \end{aligned}$$

2 Consistency relations

2.1 $n = 0$

For $p = 3$, the absolute cutoff is $W = 4$:

$$\alpha_0^t = \sum_{d=1}^{4} R_{0,d} + O(\mathrm{SR}^5).$$

Absolute order 1.

$$R_{0,1} = -0.125r$$

Absolute order 2.

$$R_{0,2} = 0.0078125r^2 + 0.125\alpha_0^s r$$

Absolute order 3.

$$\begin{aligned} R_{0,3} = &-0.0030712246746565r^3 + 0.125\alpha_1^s r \\ &- 0.10280837917801\left(\alpha_0^s\right)^2 r - 0.041327094794504\alpha_0^s r^2 \end{aligned}$$

Absolute order 4.

$$\begin{aligned} R_{0,4} = &\, 0.0014926374043055r^4 + 0.125\alpha_2^s r \\ &+ 0.065244100954277\left(\alpha_0^s\right)^3 r + 0.024554893470954\alpha_0^s r^3 \\ &+ 0.10320649151137\left(\alpha_0^s\right)^2 r^2 - 0.060109233906372\alpha_1^s r^2 \\ &- 0.23087387125098\alpha_0^s \alpha_1^s r \end{aligned}$$

2.2 $n = 1$

For $p = 3$, the absolute cutoff is $W = 5$:

$$\alpha_1^t = \sum_{d=2}^{5} R_{1,d} + O(\mathrm{SR}^6).$$

Absolute order 2.

$$R_{1,2} = 0.015625r^2 + 0.125\alpha_0^s r$$

Absolute order 3.

$$\begin{aligned} R_{1,3} = &-0.0029296875r^3 + 0.125\alpha_1^s r \\ &- 0.125\left(\alpha_0^s\right)^2 r - 0.046875\alpha_0^s r^2 \end{aligned}$$

Absolute order 4.

$$\begin{aligned} R_{1,4} = &\, 0.0016576826498282r^4 + 0.125\alpha_2^s r \\ &+ 0.12398128438351\left(\alpha_0^s\right)^2 r^2 + 0.10280837917801\left(\alpha_0^s\right)^3 r \\ &+ 0.027641022071908\alpha_0^s r^3 - 0.072577094794504\alpha_1^s r^2 \\ &- 0.33061675835603\alpha_0^s \alpha_1^s r \end{aligned}$$

Absolute order 5.

$$\begin{aligned}R_{1,5} = &-0.0010528680915447r^5 + 0.125\alpha_3^s r \\ &+ 0.050025543685844\alpha_1^s r^3 - 0.091359233906372\alpha_2^s r^2 \\ &- 0.065244100954277\left(\alpha_0^s\right)^4 r - 0.13027434666455\left(\alpha_0^s\right)^2 r^3 \\ &- 0.021075080402355\alpha_0^s r^4 - 0.23087387125098\left(\alpha_1^s\right)^2 r \\ &- 0.24842610305581\left(\alpha_0^s\right)^3 r^2 + 0.42660617411381\left(\alpha_0^s\right)^2 \alpha_1^s r \\ &+ 0.41559991864823\alpha_0^s\alpha_1^s r^2 - 0.35587387125098\alpha_0^s\alpha_2^s r\end{aligned}$$

2.3 $n = 2$

For $p = 3$, the absolute cutoff is $W = 6$:

$$\alpha_2^t = \sum_{d=3}^{6} R_{2,d} + O(\mathrm{SR}^7).$$

Absolute order 3.

$$\begin{aligned}R_{2,3} = &-0.00390625r^3 + 0.125\alpha_1^s r \\ &- 0.125\left(\alpha_0^s\right)^2 r - 0.046875\alpha_0^s r^2\end{aligned}$$

Absolute order 4.

$$\begin{aligned}R_{2,4} = &\ 0.0013427734375r^4 + 0.125\alpha_2^s r \\ &+ 0.125\left(\alpha_0^s\right)^3 r + 0.125\left(\alpha_0^s\right)^2 r^2 \\ &+ 0.025390625\alpha_0^s r^3 - 0.0625\alpha_1^s r^2 \\ &- 0.375\alpha_0^s\alpha_1^s r\end{aligned}$$

Absolute order 5.

$$\begin{aligned}R_{2,5} = &-0.00099348164677838r^5 + 0.125\alpha_3^s r \\ &+ 0.050668108270534\alpha_1^s r^3 - 0.13499234851023\left(\alpha_0^s\right)^2 r^3 \\ &- 0.28928966356152\left(\alpha_0^s\right)^3 r^2 - 0.020502543360439\alpha_0^s r^4 \\ &- 0.10280837917801\left(\alpha_0^s\right)^4 r - 0.33061675835603\left(\alpha_1^s\right)^2 r \\ &- 0.088202094794504\alpha_2^s r^2 + 0.63904189589007\left(\alpha_0^s\right)^2 \alpha_1^s r \\ &+ 0.46569385315053\alpha_0^s\alpha_1^s r^2 - 0.45561675835603\alpha_0^s\alpha_2^s r\end{aligned}$$

Absolute order 6.

$$\begin{aligned}R_{2,6} = &\ 0.0007834832445242r^6 + 0.125\alpha_4^s r \\ &+ 0.065244100954277\left(\alpha_0^s\right)^5 r + 0.51447193930681\left(\alpha_0^s\right)^3 r^3 \\ &+ 0.018883632267472\alpha_0^s r^5 + 0.077748164662437\alpha_2^s r^3 \\ &+ 0.5388653261447\left(\alpha_0^s\right)^4 r^2 + 0.15691226448912\left(\alpha_0^s\right)^2 r^4 \\ &+ 0.4600841525546\left(\alpha_1^s\right)^2 r^2 - 0.10698423390637\alpha_3^s r^2 \\ &- 0.044329625847117\alpha_1^s r^4 + 0.78248004536479\left(\alpha_0^s\right)^2 \alpha_2^s r \\ &+ 1.0840862194786\alpha_0^s\left(\alpha_1^s\right)^2 r + 0.67405262036735\alpha_0^s\alpha_2^s r^2 \\ &- 0.56686562073618\alpha_0^s\alpha_1^s r^3 - 0.68758257793092\left(\alpha_0^s\right)^3 \alpha_1^s r \\ &- 0.81762161375293\alpha_1^s\alpha_2^s r - 1.7284662943263\left(\alpha_0^s\right)^2 \alpha_1^s r^2 \\ &- 0.48087387125098\alpha_0^s\alpha_3^s r\end{aligned}$$

2.4 $n = 3$

For $p = 3$, the absolute cutoff is $W = 7$:

$$\alpha_3^t = \sum_{d=4}^{7} R_{3,d} + O(\mathrm{SR}^8).$$

Absolute order 4.

$$\begin{aligned} R_{3,4} = {} & 0.00146484375 r^4 + 0.125 \alpha_2^s r \\ & + 0.125 \left(\alpha_0^s\right)^3 r + 0.0234375 \alpha_0^s r^3 \\ & + 0.109375 \left(\alpha_0^s\right)^2 r^2 - 0.0625 \alpha_1^s r^2 \\ & - 0.375 \alpha_0^s \alpha_1^s r \end{aligned}$$

Absolute order 5.

$$\begin{aligned} R_{3,5} = {} & -0.000762939453125 r^5 + 0.125 \alpha_3^s r \\ & + 0.0419921875 \alpha_1^s r^3 - 0.125 \left(\alpha_0^s\right)^4 r \\ & - 0.375 \left(\alpha_1^s\right)^2 r - 0.078125 \alpha_2^s r^2 \\ & - 0.28125 \left(\alpha_0^s\right)^3 r^2 - 0.01708984375 \alpha_0^s r^4 \\ & - 0.1201171875 \left(\alpha_0^s\right)^2 r^3 + 0.75 \left(\alpha_0^s\right)^2 \alpha_1^s r \\ & + 0.4375 \alpha_0^s \alpha_1^s r^2 - 0.5 \alpha_0^s \alpha_2^s r \end{aligned}$$

Absolute order 6.

$$\begin{aligned} R_{3,6} = {} & 0.00069887861331449 r^6 + 0.125 \alpha_4^s r \\ & + 0.52264594794504 \left(\alpha_1^s\right)^2 r^2 + 0.10280837917801 \left(\alpha_0^s\right)^5 r \\ & + 0.52433019631832 \left(\alpha_0^s\right)^3 r^3 + 0.017257388611671 \alpha_0^s r^5 \\ & + 0.07369519446916 \alpha_2^s r^3 + 0.1495700174229 \left(\alpha_0^s\right)^2 r^4 \\ & + 0.61990642191755 \left(\alpha_0^s\right)^4 r^2 - 0.042328393296221 \alpha_1^s r^4 \\ & - 0.1038270947945 \alpha_3^s r^2 + 1.6087005501362 \alpha_0^s \left(\alpha_1^s\right)^2 r \\ & + 1.0946586542461 \left(\alpha_0^s\right)^2 \alpha_2^s r + 0.71467513753404 \alpha_0^s \alpha_2^s r^2 \\ & - 0.57385180946901 \alpha_0^s \alpha_1^s r^3 - 1.9544879813691 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \\ & - 1.1168502750681 \alpha_1^s \alpha_2^s r - 1.0502754126021 \left(\alpha_0^s\right)^3 \alpha_1^s r \\ & - 0.58061675835603 \alpha_0^s \alpha_3^s r \end{aligned}$$

Absolute order 7.

$$\begin{aligned} R_{3,7} = {} & -0.00066040799045703 r^7 + 0.125 \alpha_5^s r \\ & + 0.044182253542082 \alpha_1^s r^5 + 0.10547078563903 \alpha_3^s r^3 \\ & + 1.0840862194786 \left(\alpha_1^s\right)^3 r - 0.12260923390637 \alpha_4^s r^2 \\ & - 0.065244100954277 \left(\alpha_0^s\right)^6 r - 1.8011431833565 \left(\alpha_0^s\right)^4 r^3 \\ & - 1.1197437723225 \left(\alpha_0^s\right)^5 r^2 - 0.076712092161027 \alpha_2^s r^4 \\ & - 0.70760825653778 \left(\alpha_1^s\right)^2 r^3 - 0.19445546235151 \left(\alpha_0^s\right)^2 r^5 \\ & - 0.018648400200653 \alpha_0^s r^6 - 0.81762161375293 \left(\alpha_2^s\right)^2 r \\ & - 0.9077746502428 \left(\alpha_0^s\right)^3 r^4 + 0.75724388321833 \alpha_0^s \alpha_1^s r^4 \\ & + 1.2633539166158 \left(\alpha_0^s\right)^2 \alpha_3^s r + 5.8694148411763 \left(\alpha_0^s\right)^3 \alpha_1^s r^2 \\ & + 3.899202830862 \left(\alpha_0^s\right)^2 \alpha_1^s r^3 + 1.0138030827023 \left(\alpha_0^s\right)^4 \alpha_1^s r \end{aligned}$$

$$
\begin{aligned}
&+ 1.7276736271957\alpha_1^s\alpha_2^s r^2 + 0.96375532208646\alpha_0^s\alpha_3^s r^2 \\
&- 1.0111179362879\alpha_0^s\alpha_2^s r^3 - 0.60587387125098\alpha_0^s\alpha_4^s r \\
&- 3.1468339532713\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r \\
&- 4.6296729998974\alpha_0^s\left(\alpha_1^s\right)^2 r^2 - 3.2598096829233\left(\alpha_0^s\right)^2\alpha_2^s r^2 \\
&- 1.2984954850039\alpha_1^s\alpha_3^s r - 1.4700626232957\left(\alpha_0^s\right)^3\alpha_2^s r \\
&+ 4.5507541434397\alpha_0^s\alpha_1^s\alpha_2^s r
\end{aligned}
$$